

Gradient Boosting Classification Notes

Introduction to Gradient Boosting

Gradient Boosting is one of the most important Machine Learning algorithms. It is a boosting technique that combines multiple weak models step by step to create a strong predictive model.

The main idea is:

- Train a simple model first
- Find its mistakes
- Pass those mistakes to the next model
- The next model tries to correct previous errors
- Repeat this process multiple times

This process is called **Additive Modeling** because models are added stage by stage.

Gradient Boosting works for:

- Regression Problems
- Classification Problems

The overall concept remains the same in both cases. The main difference is the **Loss Function** used.

Core Idea of Boosting

Boosting combines multiple weak learners.

A weak learner:

- Performs slightly better than random guessing
- Usually has high bias and low variance

Gradient Boosting reduces bias by combining many weak learners together.

The algorithm trains models sequentially:

- Model 1 learns from data
- Model 2 learns from Model 1 errors
- Model 3 learns from combined errors
- And so on

Each new model improves overall performance.

Gradient Boosting for Classification

In classification problems:

- We use **Log Loss** as the loss function
- Small Decision Trees are used as weak learners
- Trees are added sequentially

The algorithm does not change completely from regression. Only the loss function changes.

Example Dataset

Suppose we have a placement prediction dataset with:

Input Features

- CGPA
- IQ

Output

- Placement (Yes/No)

The goal is to predict whether a student will get placement.

Stage 1: Initial Model

The first model in Gradient Boosting is very simple.

For regression:

- Mean value is used

For classification:

- Log Odds are used

Log Odds Formula

$$\log \left(\frac{p}{1-p} \right)$$

Where:

- p = probability of positive class

Converting Log Odds to Probability

Probability can be calculated using the sigmoid function.

$$p = \frac{e^z}{1+e^z}$$

Where:

- z = log odds value

Initial Prediction

The first model predicts the same probability for all data points.

Example:

- If most students are placed
- The model predicts placement for everyone

This is not accurate, but it is a good starting point.

Calculating Residuals

After predictions are made, errors are calculated.

These errors are called **Pseudo Residuals**.

Formula:

$$\text{Residual} = \text{Actual} - \text{Prediction}$$

Residuals tell us:

- How wrong the model is
- What the next model should learn

Stage 2: Building the First Decision Tree

A small Decision Tree is trained using:

Inputs

- CGPA
- IQ

Target

- Residuals

Important:

- Even in classification, a **Regression Tree** is used
- Trees are kept shallow to remain weak learners

Leaf Output Calculation

For every leaf node, a special value called Gamma is calculated.

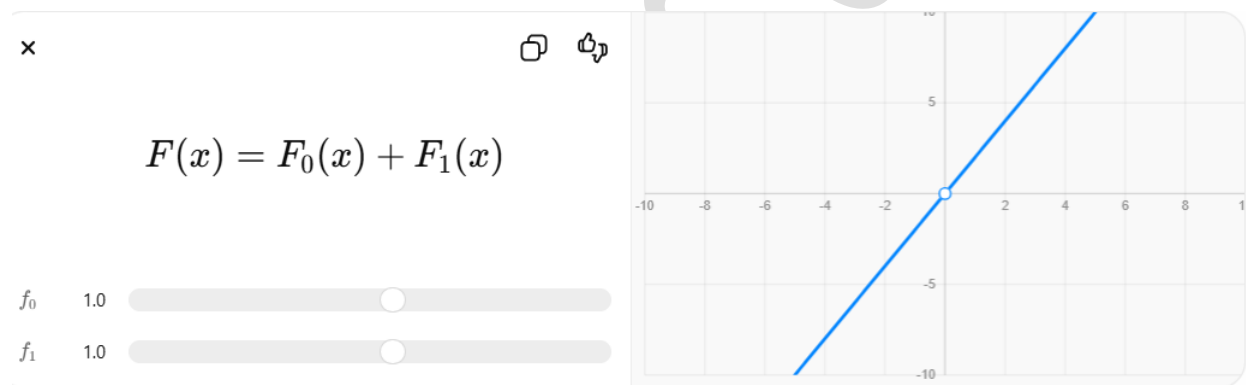
Formula:

$$\gamma = \frac{\sum residuals}{\sum p(1-p)}$$

This value helps update predictions.

Updating Predictions

The combined prediction becomes:



Where:

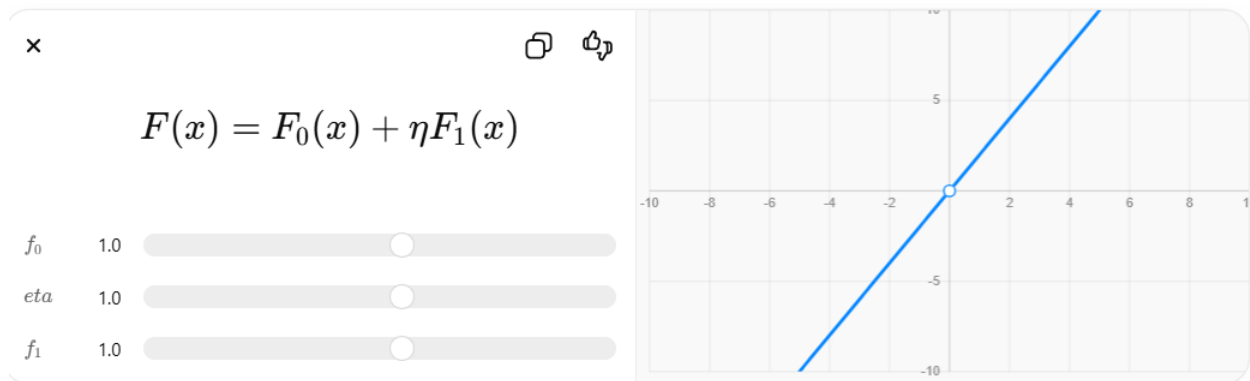
- $F_0(x)$ = initial model
- $F_1(x)$ = first decision tree

Predictions are again converted into probabilities.

Learning Rate

Gradient Boosting uses a Learning Rate to control updates.

Instead of adding full tree output:



Where:

- η = learning rate

Usually:

- $\eta=0.1$

Benefits:

- Prevents sudden large changes
- Makes learning gradual
- Reduces overfitting

Stage 3 and Beyond

The process repeats again:

1. Calculate new residuals
2. Train another tree
3. Add tree output to existing model
4. Update predictions

Final model:

$$F(x) = F_0(x) + F_1(x) + F_2(x) + \dots + F_n(x)$$

Each new tree reduces prediction errors further.

Geometric Intuition

The video also explains Gradient Boosting visually.

Initial Boundary

- First model creates a very simple decision boundary

After Adding Trees

- Decision boundaries become more flexible
- Model learns complex patterns
- Residuals reduce gradually

Each new tree slightly adjusts the boundary to improve classification accuracy.

Important Concepts

Weak Learners

- Small decision trees
- High bias models
- Simple and fast

Additive Modeling

- Models are added sequentially

Residual Learning

- Every new model learns previous errors

Learning Rate

- Controls contribution of each tree

Log Loss

- Used in classification problems

Key Advantages of Gradient Boosting

- High prediction accuracy
- Works for both regression and classification
- Handles complex patterns
- Reduces bias effectively
- Very powerful ensemble technique

Final Summary

Gradient Boosting Classification works by:

1. Starting with a simple prediction model
2. Calculating errors (residuals)
3. Training small trees on those errors
4. Sequentially improving predictions
5. Combining all weak learners into one strong model

The algorithm gradually improves performance and creates highly accurate classification models.